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Supplemental information

Solvent tuning regulates proton flux to extend stability in reactive CO_2 capture and electrolysis

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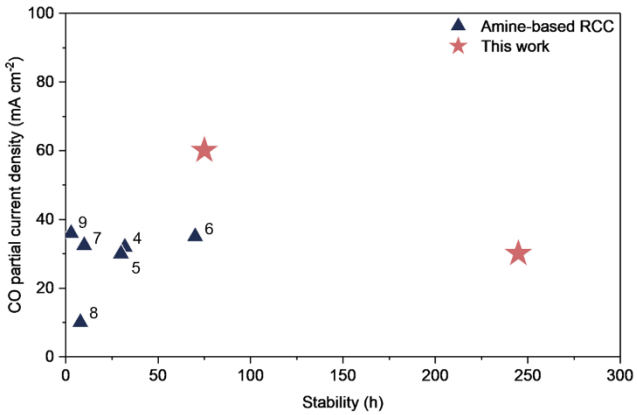

Figure S1. Summary of state-of-the-art amine-based RCC, comparing their CO partial current density and stability.[1–6] Details are tabulated in Table S1.

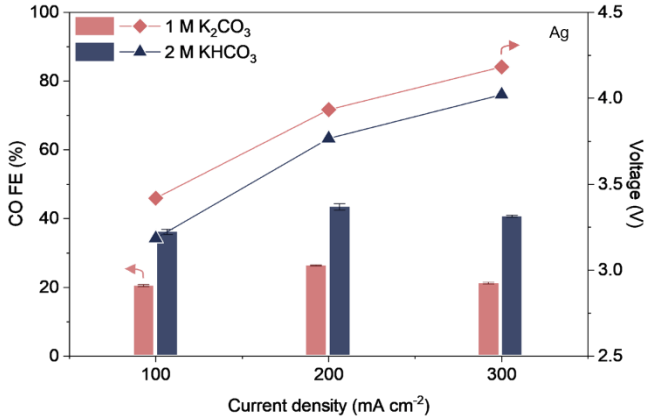

Figure S2. CO Faradaic efficiency (FE) and full cell operating voltage at current densities ranging from 100 mA cm^{-2} to 300 mA cm^{-2} using an Ag catalyst and 1 M K_2CO_3 or 2 M $KHCO_3$ as the catholyte. The catholyte was not saturated with CO_2.

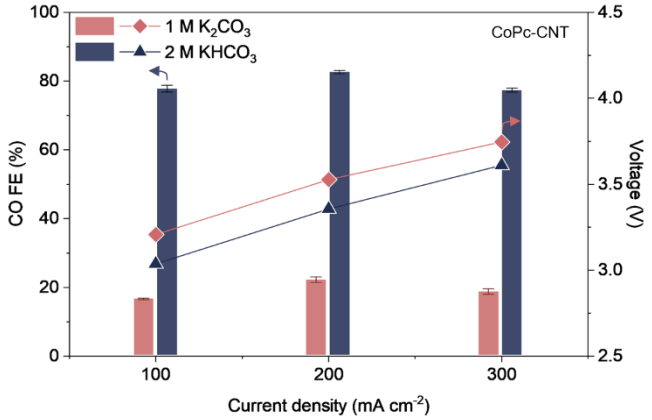

Figure S3. CO FE and full cell operating voltage at current densities ranging from 100 mA cm^{-2} to 300 mA cm^{-2} using a baseline CoPc-CNT catalyst and 1 M K_2CO_3 or 2 M $KHCO_3$ as the catholyte. The catholyte was not saturated with CO_2.

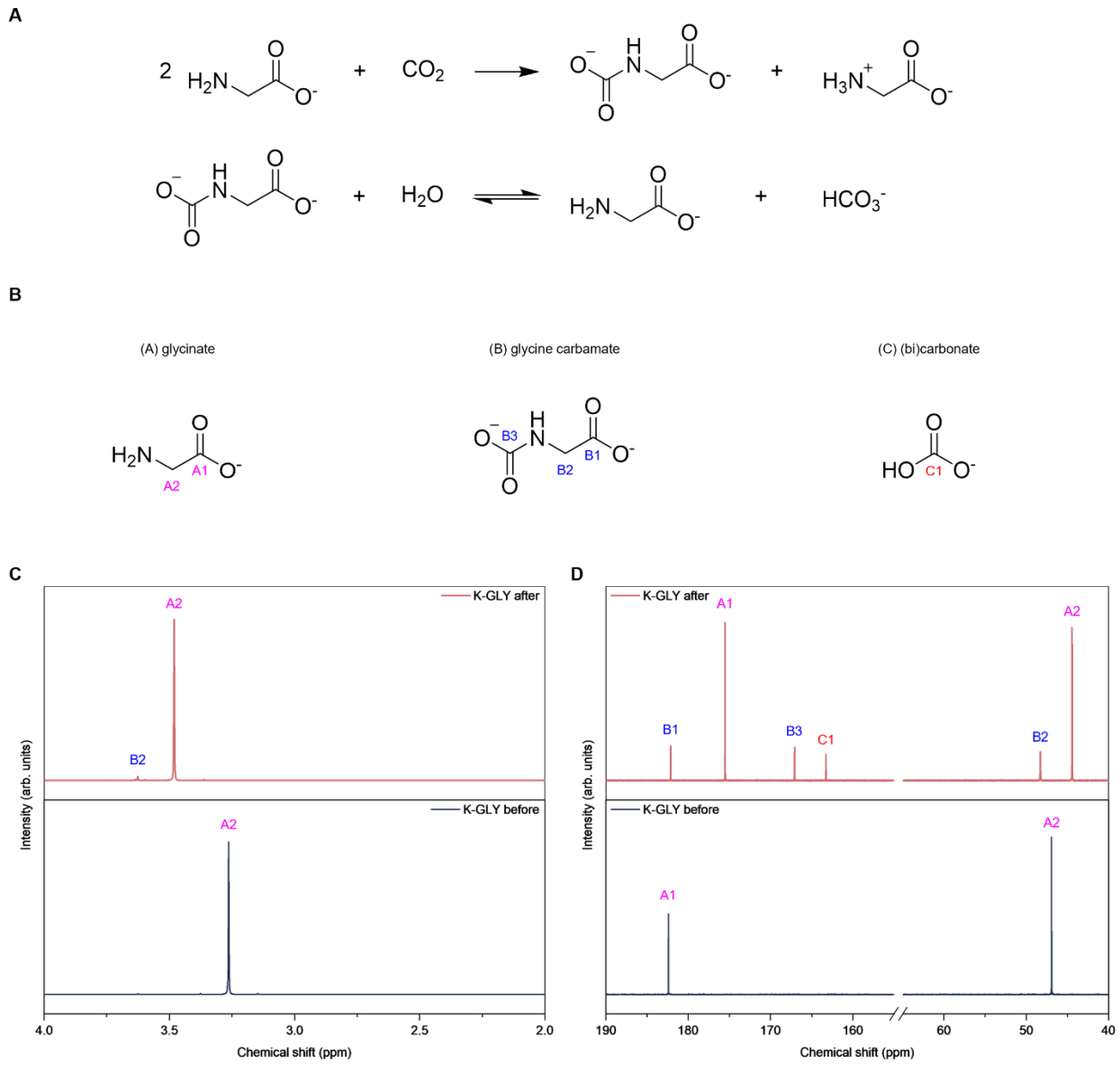

Figure S4. Chemical species before and after CO_2 capture using a 2 M potassium glycinate (K-GLY) solvent. (A) Chemical equations for the glycinate–CO_2 reaction leading to the formation of glycine carbamate and glycine zwitterion, and the hydrolysis of glycine carbamate to form bicarbonate and regenerate glycinate. (B) Chemical structures of glycinate, glycine carbamate, and bicarbonate. (C and D) ^{1}H NMR and ^{13}C NMR spectra before and after CO_2 capture with peaks labelled.

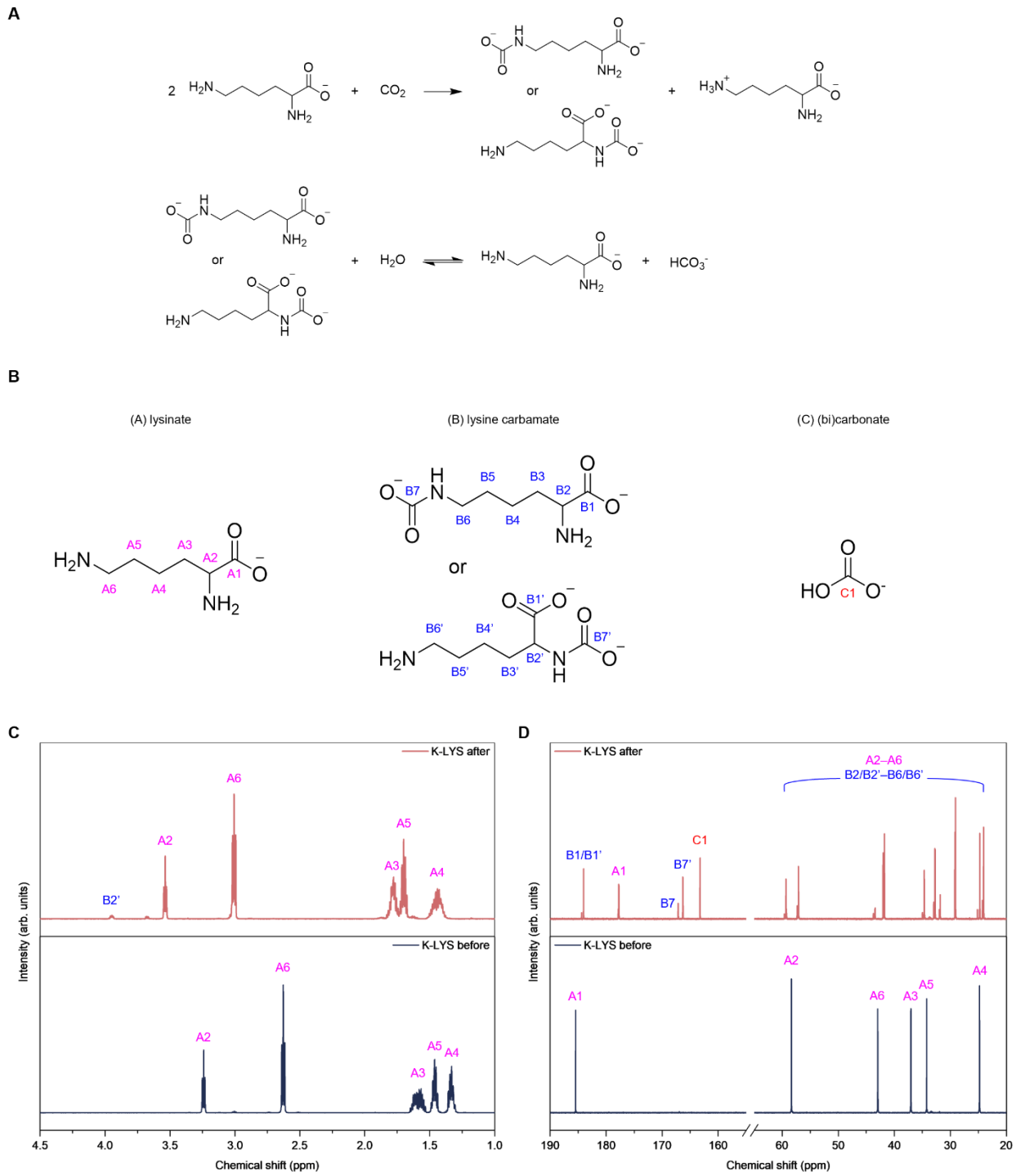

Figure S5. Chemical species before and after CO_2 capture using a 2 M potassium lysinate (K-LYS) solvent. (A) Chemical equations for the lysinate–CO_2 reaction leading to the formation of lysine carbamate and lysine zwitterion, and the hydrolysis of lysine carbamate to form bicarbonate and regenerate lysinate. (B)

Chemical structures of lysinate, lysine carbamate, and bicarbonate. (C and D) ^{1}H NMR and ^{13}C NMR spectra before and after CO_2 capture with peaks labelled.

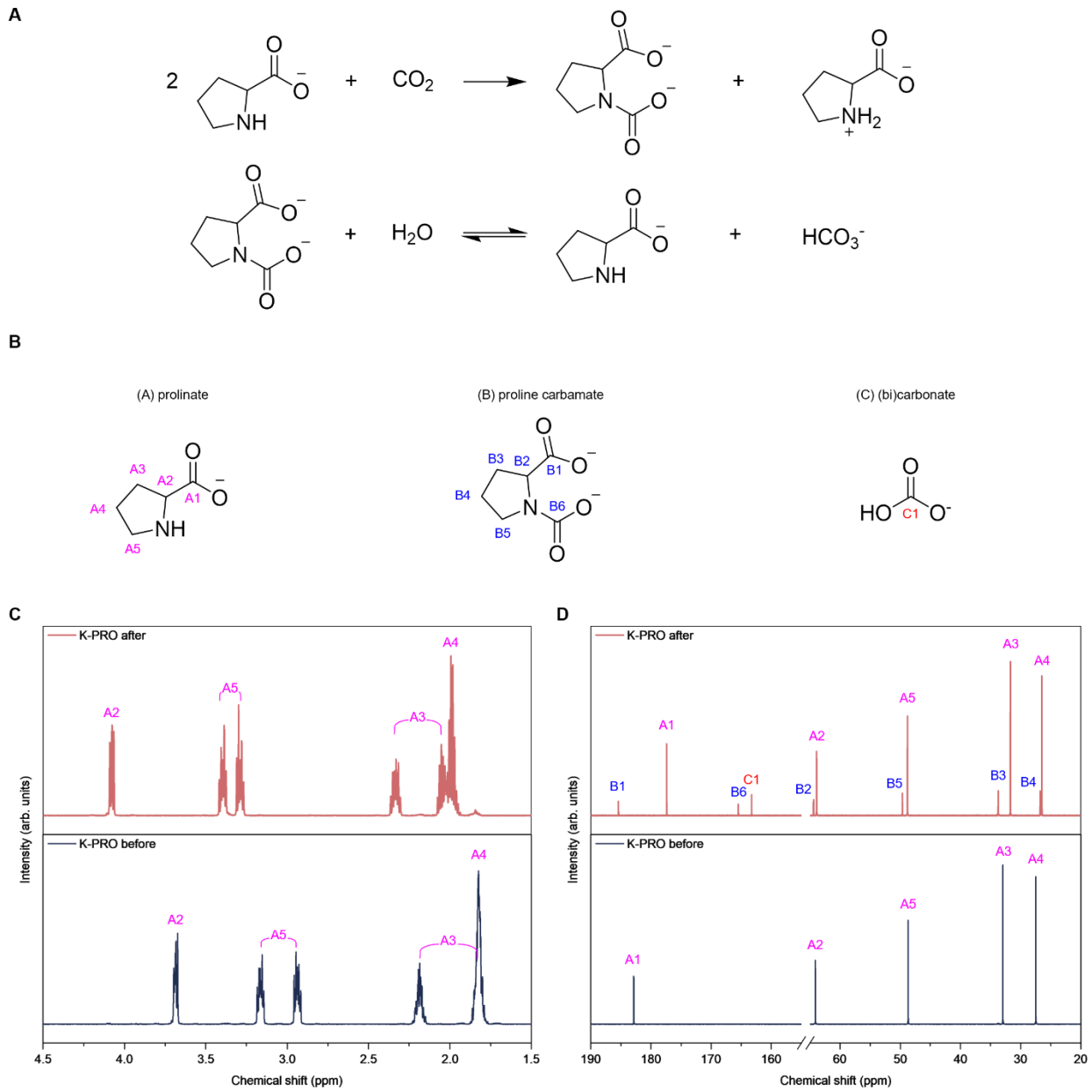

Figure S6. Chemical species before and after CO_2 capture using a 2 M potassium prolinate (K-PRO) solvent. (A) Chemical equations for the prolinate–CO_2 reaction leading to the formation of proline carbamate and proline zwitterion, and the hydrolysis of proline carbamate to form bicarbonate and regenerate prolinate. (B) Chemical structures of prolinate, proline carbamate, and bicarbonate. (C and D) ^{1}H NMR and ^{13}C NMR spectra before and after CO_2 capture with peaks labelled.

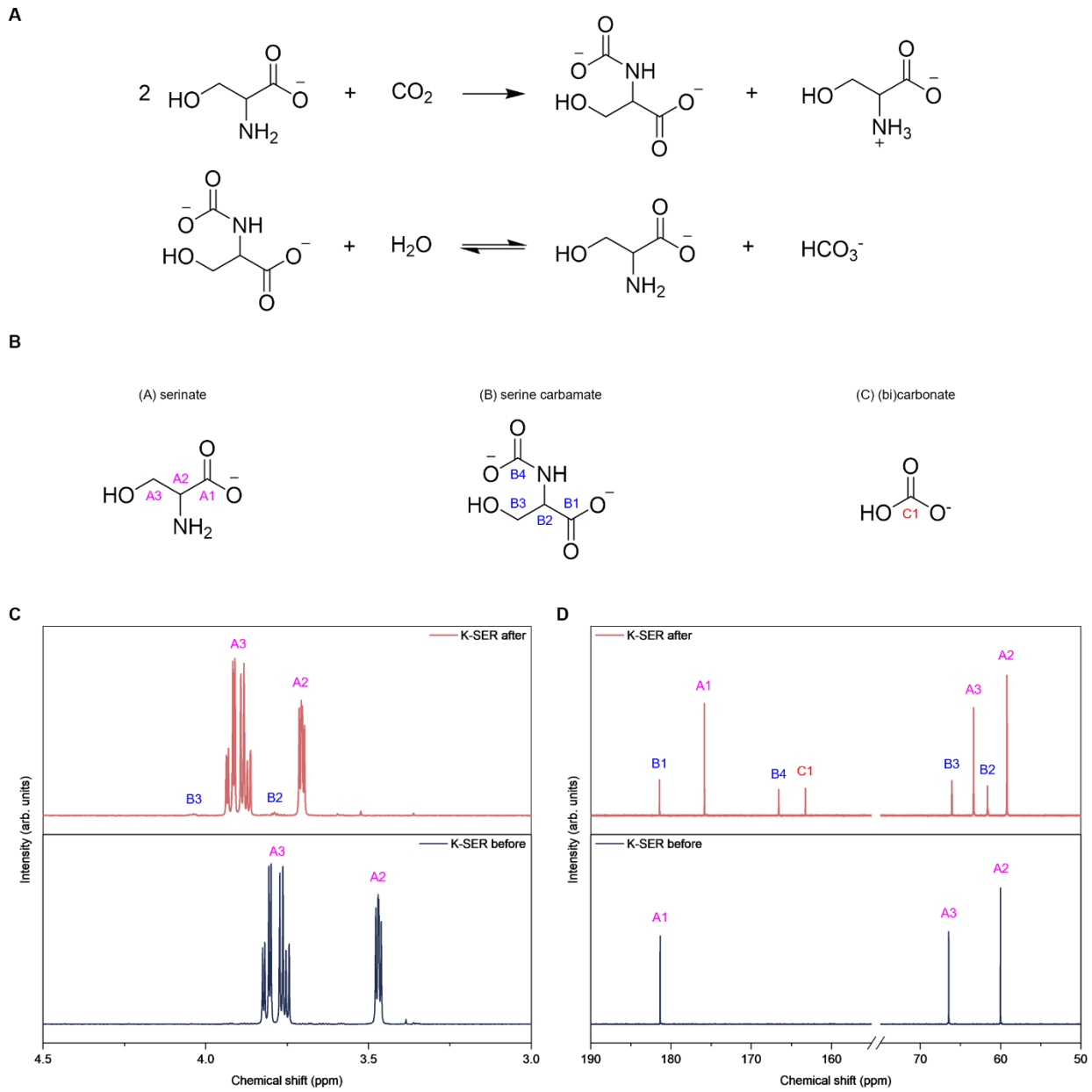

Figure S7. Chemical species before and after CO_2 capture using a 2 M potassium serinate (K-SER) solvent. (A) Chemical equations for the serinate–CO_2 reaction leading to the formation of serine carbamate and serine zwitterion, and the hydrolysis of serine carbamate to form bicarbonate and regenerate serinate. (B) Chemical structures of serinate, serine carbamate, and bicarbonate. (C and D) ^{1}H NMR and ^{13}C NMR spectra before and after CO_2 capture with peaks labelled.

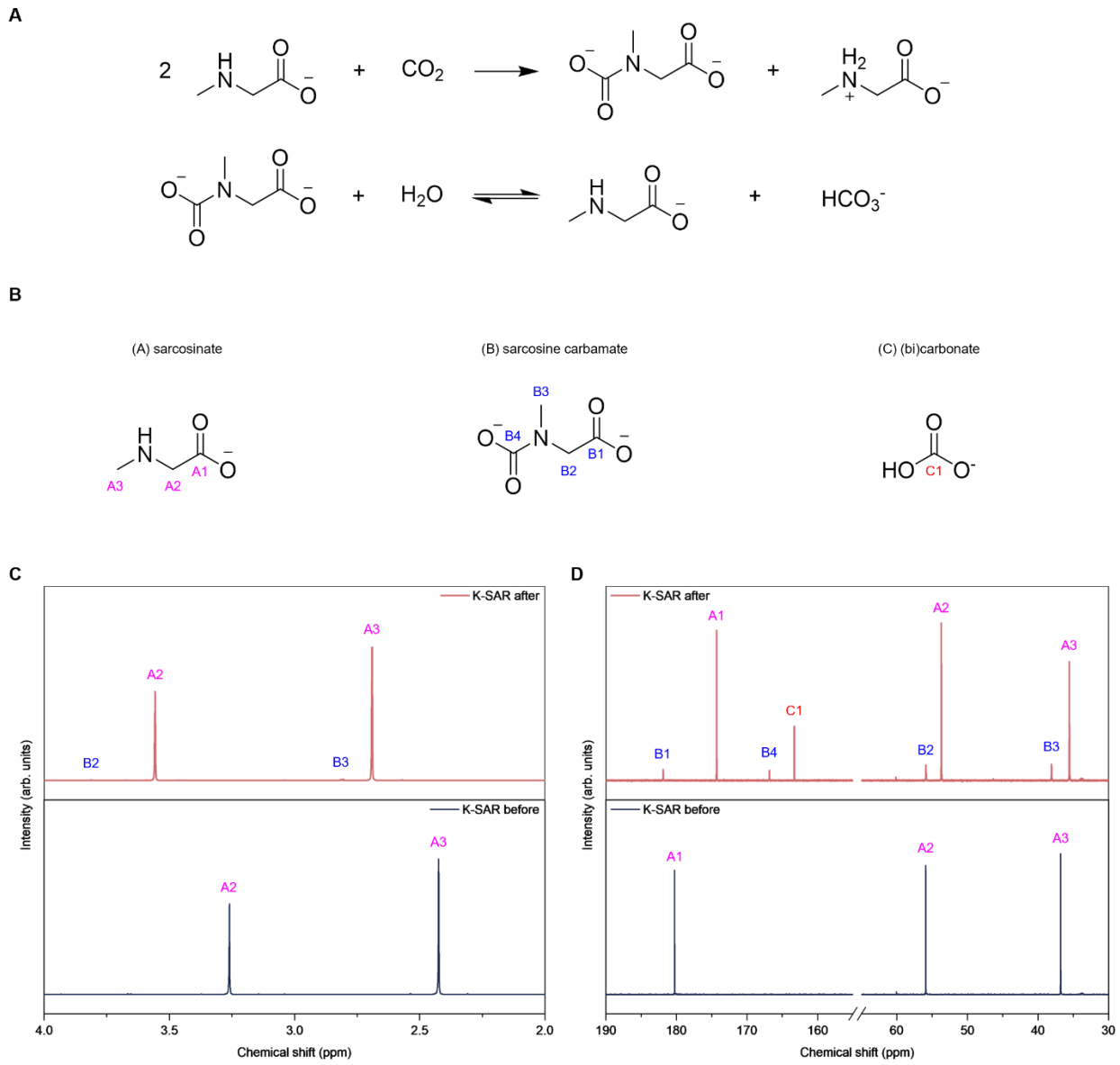

Figure S8. Chemical species before and after CO_2 capture using a 2 M potassium sarcosinate (K-SAR) solvent. (A) Chemical equations for the sarcosinate–CO_2 reaction leading to the formation of sarcosine carbamate and sarcosine zwitterion, and the hydrolysis of sarcosine carbamate to form bicarbonate and regenerate sarcosinate. (B) Chemical structures of sarcosinate, sarcosine carbamate, and bicarbonate. (C and D) ^{1}H NMR and ^{13}C NMR spectra before and after CO_2 capture with peaks labelled.

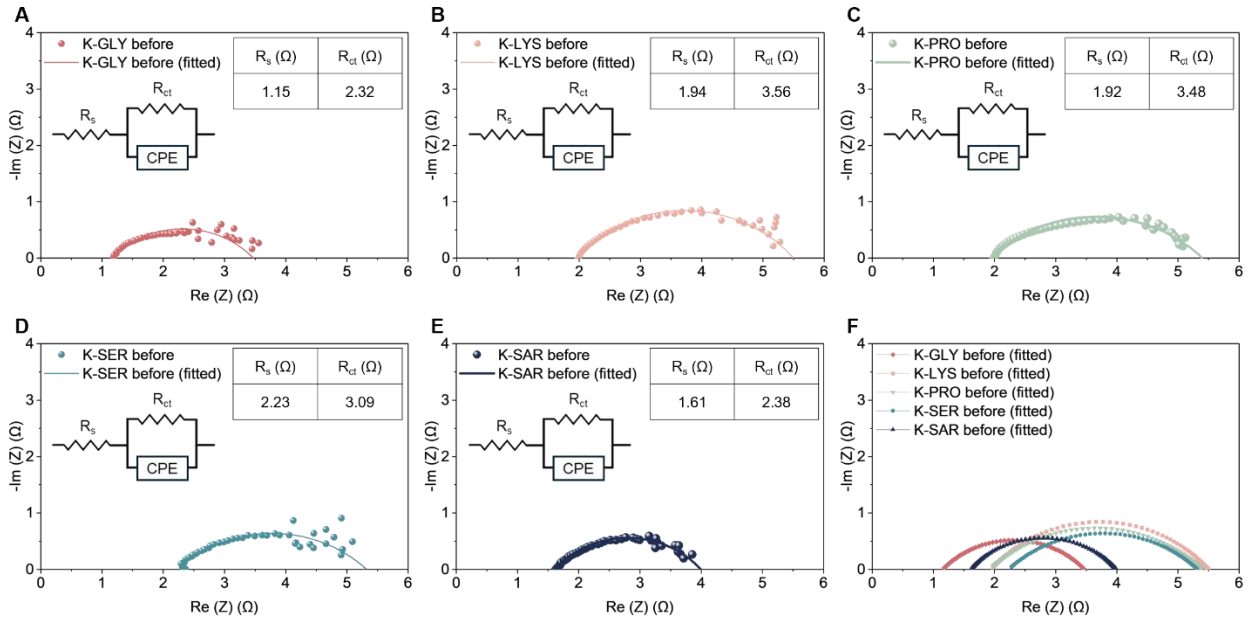

Figure S9. Full cell EIS of RCC before prolonged electrolysis using various amino acid salt capture solvents. (A) K-GLY. (B) K-LYS. (C) K-PRO. (D) K-SER. (E) K-SAR. (F) Fitted circuits of all amino acid salt solvents. The spectra were obtained at 100 mA cm^{-2}, with an electrode area of 1 cm^2, after 300 seconds of system equilibration using a baseline CoPc-CNT catalyst. A Randles circuit with a constant phase element was used for circuit fitting and obtaining the R_s and R_{ct}.

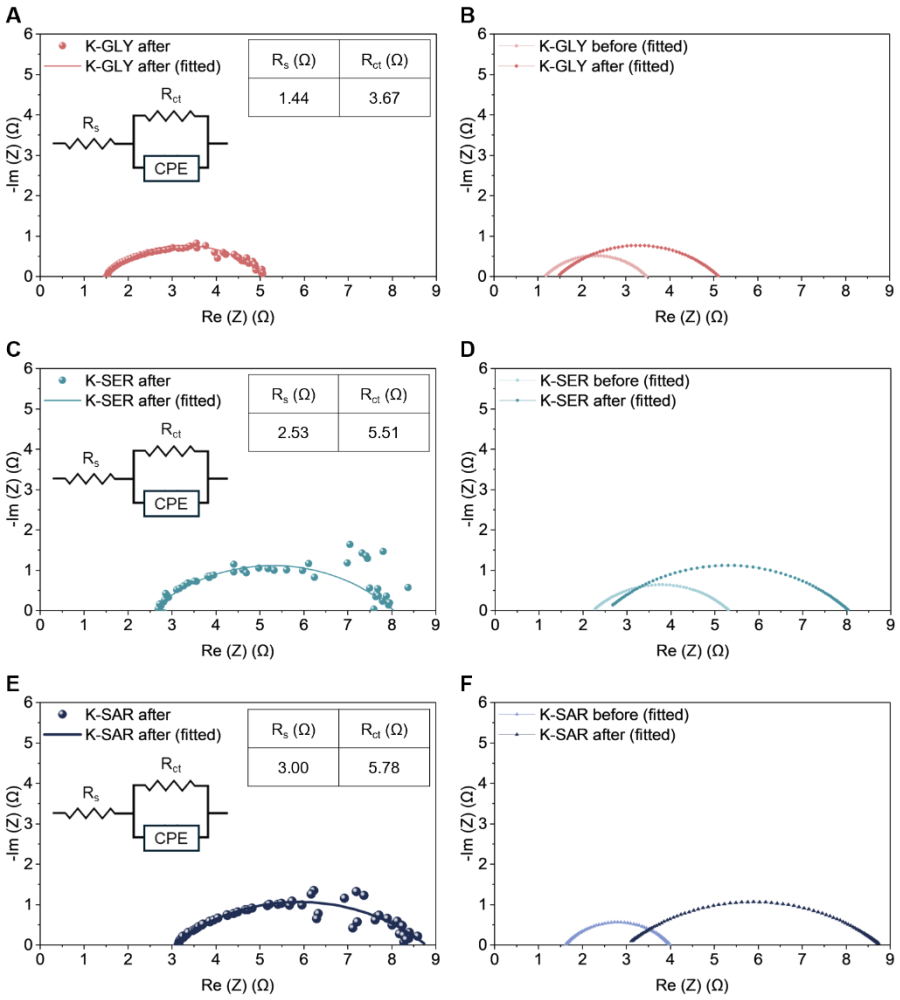

Figure S10. Full cell EIS of RCC after prolonged electrolysis using various amino acid salt capture solvents. (A) K-GLY. (B) Fitted circuits of K-GLY before and after electrolysis. (C) K-SER. (D) Fitted circuits of K-SER before and after electrolysis. (E) K-SAR. (F) Fitted circuits of K-SAR before and after electrolysis. The spectra were obtained at 100 mA cm^{-2}, with an electrode area of 1 cm^2, after the stability experiments have concluded (CO FE < 20%) using a baseline CoPc-CNT catalyst. A Randles circuit with a constant phase element was used for circuit fitting and obtaining the R_s and R_{ct}.

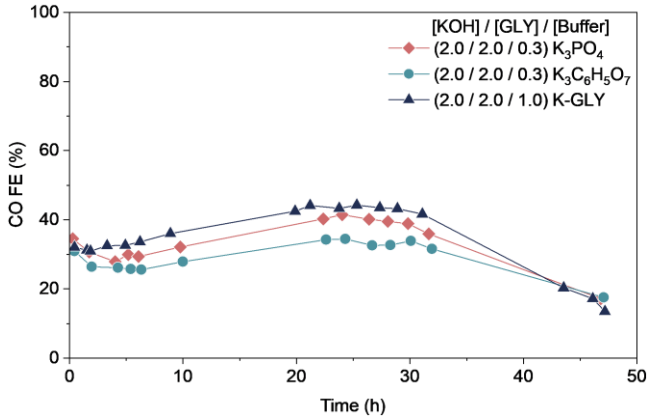

Figure S11. CO FE over time at 100 mA cm^{-2} with 2.0 M KOH + 2.0 M GLY as the capture solvent and various buffers added: 0.3 M K_3PO_4, 0.3 M $K_3C_6H_5O_7$, or 1.0 M K-GLY. The total concentration of potassium ions was controlled to 3.0 M.

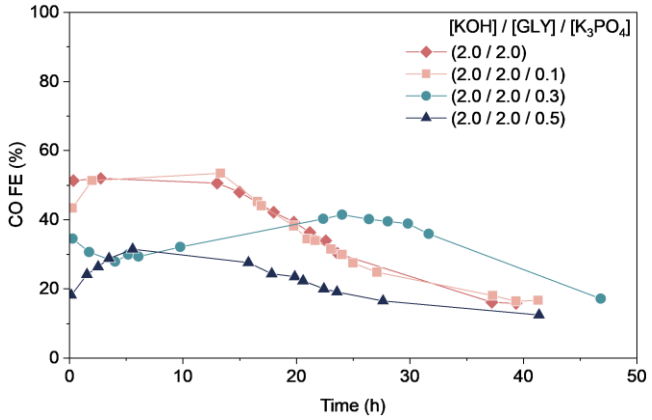

Figure S12. CO FE over time at 100 mA cm^{-2} with 2.0 M KOH + 2.0 M GLY as the capture solvent and K_3PO_4 concentrations ranging from 0 M to 0.5 M.

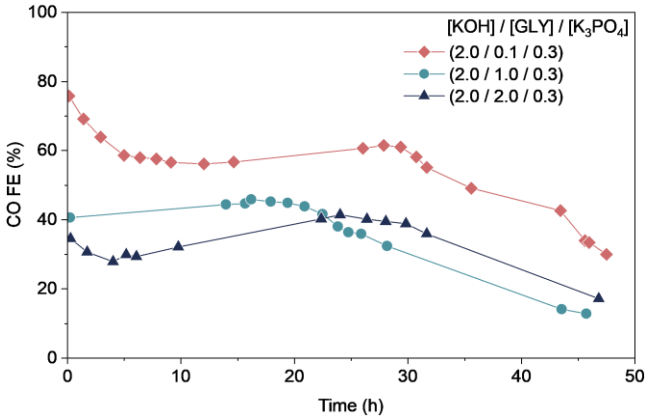

Figure S13. CO FE over time at 100 mA cm^{-2} with 2.0 M KOH and 0.3 M K_3PO_4 in the capture solvent and GLY concentrations ranging from 0.1 M to 2.0 M.

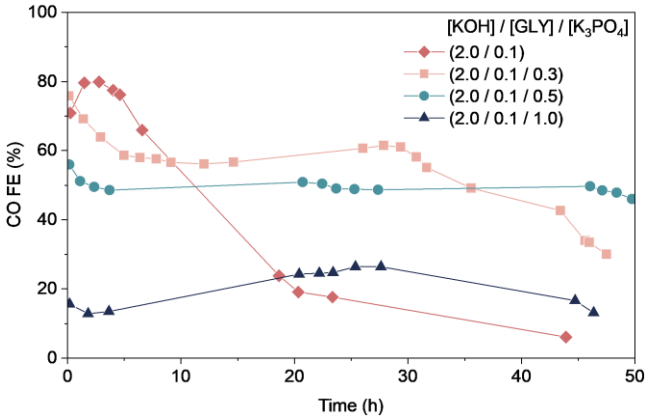

Figure S14. CO FE over time at 100 mA cm^{-2} with 2.0 M KOH + 0.1 M GLY as the capture solvent and K_3PO_4 concentrations ranging from 0 M to 1.0 M.

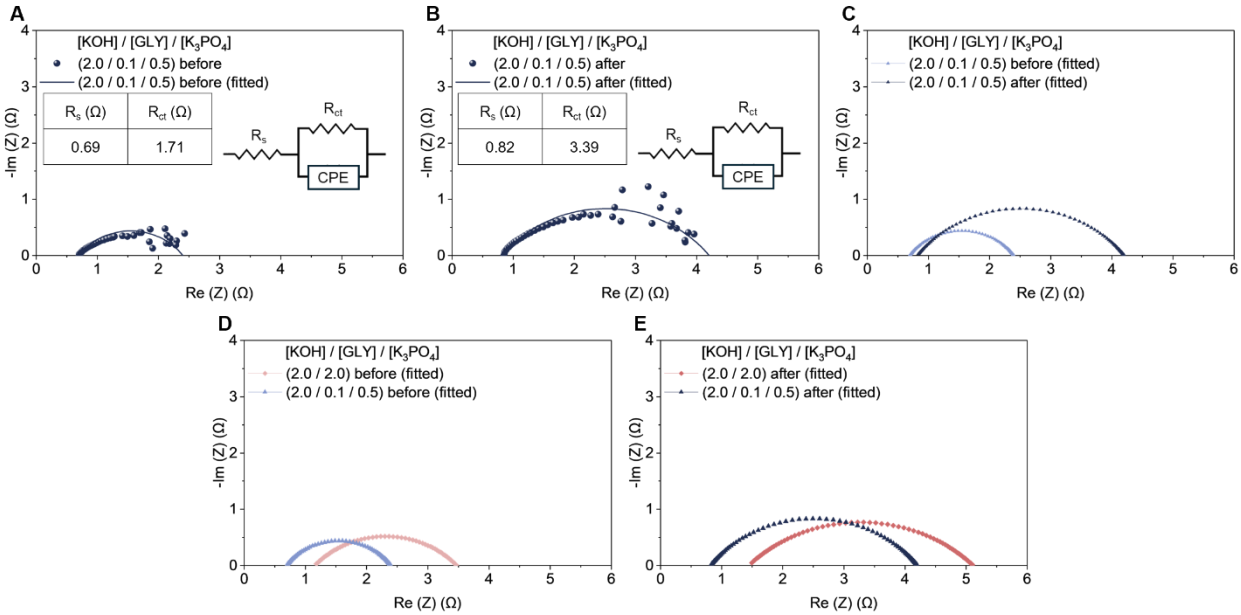

Figure S15. Comparing the full cell EIS of RCC, with a baseline CoPc-CNT catalyst, using the optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4 against the EIS using 2 M K-GLY as the capture solvent. (A) Optimized capture solvent before prolonged electrolysis. (B) Optimized capture solvent after prolonged electrolysis. (C) Fitted circuits of the optimized solvent before and after electrolysis. (D) Fitted circuits of the optimized solvent and 2 M K-GLY before electrolysis. (E) Fitted circuits of the optimized solvent and 2 M K-GLY after electrolysis. The spectra were obtained at 100 mA cm^{-2}, with an electrode area of 1 cm^2. The pre-electrolysis spectra were analyzed after 300 seconds of system equilibration and the post-electrolysis spectra were analyzed after the stability experiments have concluded (CO FE < 20%). A Randles circuit with a constant phase element was used for circuit fitting and obtaining the R_s and R_{ct}.

Figure S16. Schematic of the one-dimensional model with the thickness of each component indicated. X = 0 represents where the Distance from Cathode is calculated from in the model.

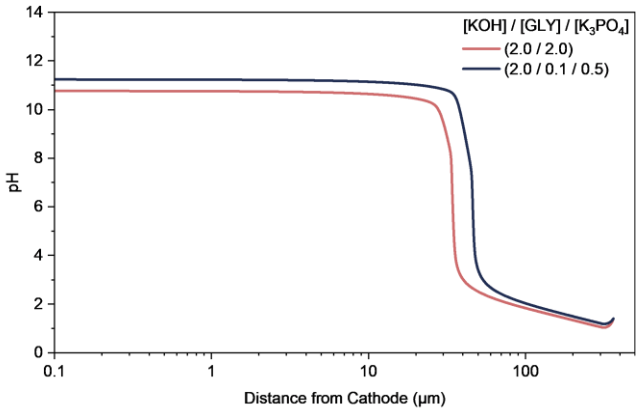

Figure S17. One-dimensional multi-physics modelling of pH at distances from the cathode at an applied current density of 200 mA cm^{-2}.

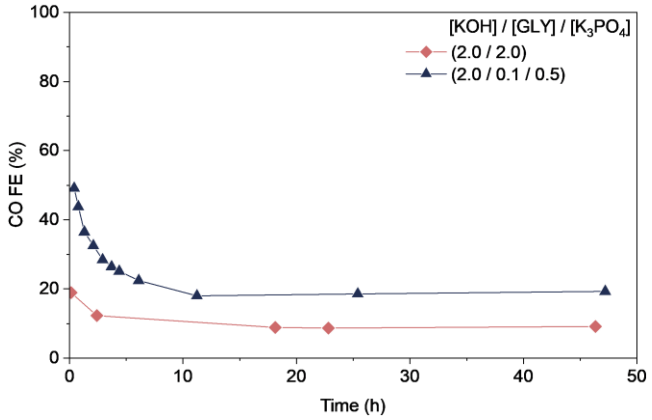

Figure S18. CO FE over time at 100 mA cm^{-2} using an Ag catalyst with 2 M K-GLY as the capture solvent and with the optimized solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4.

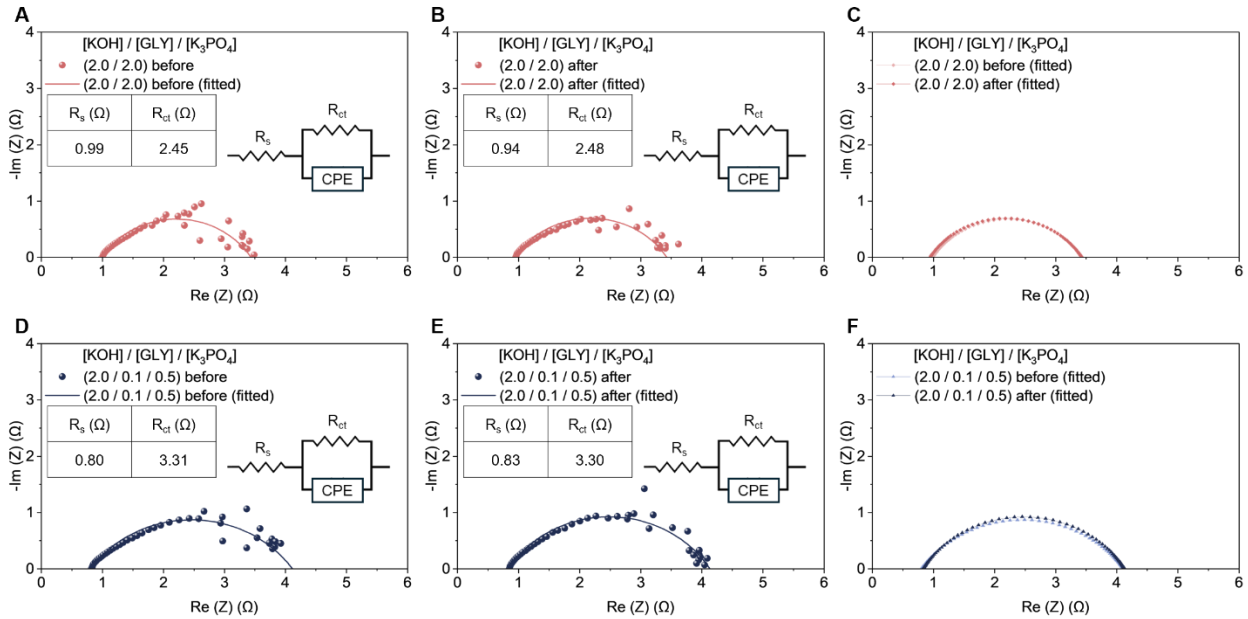

Figure S19. Comparing the full cell EIS of RCC, with an Ag catalyst, using the optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4 against the EIS using 2 M K-GLY as the capture solvent. (A) 2 M K-GLY before 50 hours of electrolysis. (B) 2 M K-GLY after 50 hours of electrolysis. (C) Fitted circuits of 2 M K-GLY before and after electrolysis. (D) Optimized capture solvent before 50 hours of electrolysis. (E) Optimized capture solvent after 50 hours of electrolysis. (F) Fitted circuits of the optimized capture solvent before and after electrolysis. The spectra were obtained at 100 mA cm^{-2}, with an electrode area of 1 cm^2. The pre-electrolysis spectra were analyzed after 300 seconds of system equilibration. A Randles circuit with a constant phase element was used for circuit fitting and obtaining the R_s and R_{ct}.

Figure S20. Picture of catholyte flow channel after 50 hours of electrolysis using a baseline CoPc-CNT catalyst.

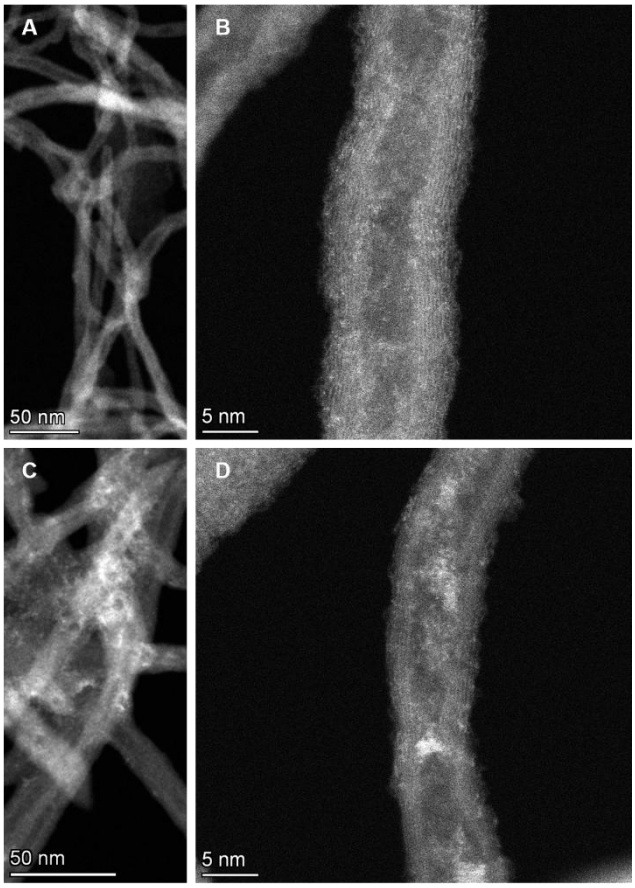

Figure S21. Scanning transmission electron microscopy (STEM) images of a baseline CoPc-CNT catalyst. (A and B) Before electrolysis. (C and D) After 50 hours of electrolysis.

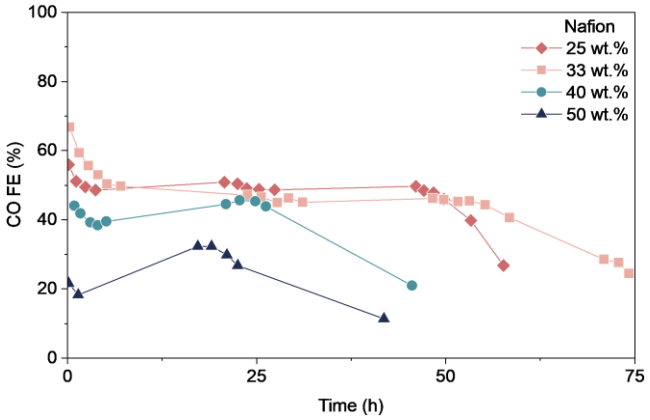

Figure S22. CO FE over time at 100 mA cm^{-2} with various Nafion content in the CoPc-CNT catalyst dispersion ranging from 25 wt.% to 50 wt.%.

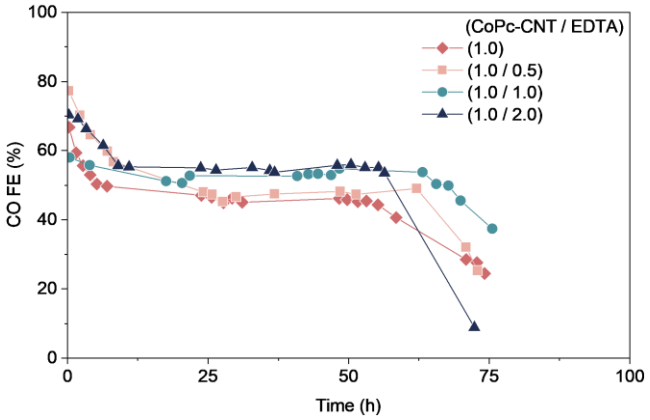

Figure S23. CO FE over time at 100 mA cm^{-2} with various CoPc-CNT:EDTA weight ratios in the catalyst dispersion ranging from 0 to 2.0.

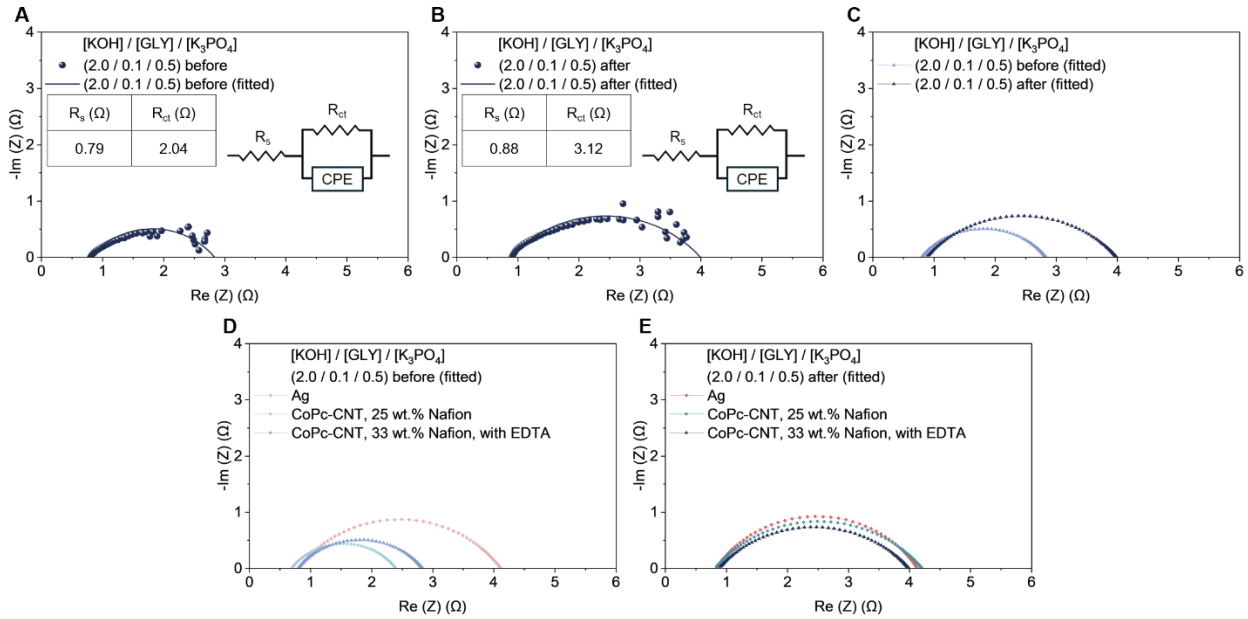

Figure S24. Comparing the full cell EIS of RCC with an Ag catalyst, a baseline CoPc-CNT catalyst, and a confined CoPc-CNT catalyst using the optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4. (A) Confined CoPc-CNT catalyst before 50 hours of electrolysis. (B) Confined CoPc-CNT catalyst after 50 hours of electrolysis. (C) Fitted circuits with confined CoPc-CNT catalyst before and after electrolysis. (D) Fitted circuits of all catalysts before 50 hours of electrolysis. (E) Fitted circuits of all catalysts after 50 hours of electrolysis. The spectra were obtained at 100 mA cm^{-2}, with an electrode area of 1 cm^2. The pre-electrolysis spectra were analyzed after 300 seconds of system equilibration. A Randles circuit with a constant phase element was used for circuit fitting and obtaining the R_s and R_{ct}.

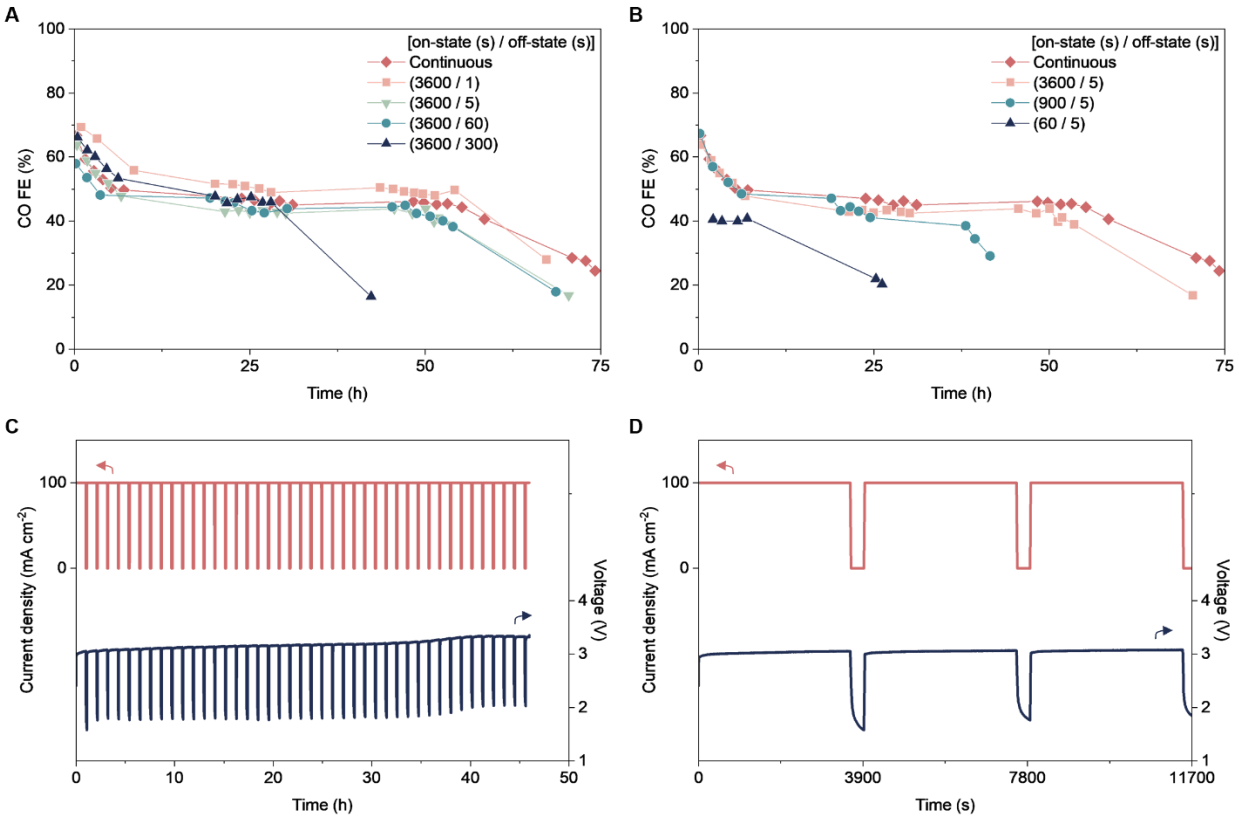

Figure S25. Electrolyzer performance over time with pulsed electrolysis between an on-state (100 mA cm^{-2}) and an off-state (0 mA cm^{-2}) at various frequencies. (A) CO FE over time with 3,600 seconds of on-state and the off-state period ranging from 0 second to 300 seconds. (B) CO FE over time with 5 seconds of off-state and on-state period ranging from 60 seconds to 3,600 seconds. (C) Current density and full cell operating voltage of pulsed electrolysis with 3,600 seconds of on-state and 300 seconds of off-state over an hourly timescale. (D) Current density and full cell operating voltage of pulsed electrolysis with 3,600 seconds of on-state and 300 seconds of off-state with seconds-scale resolution.

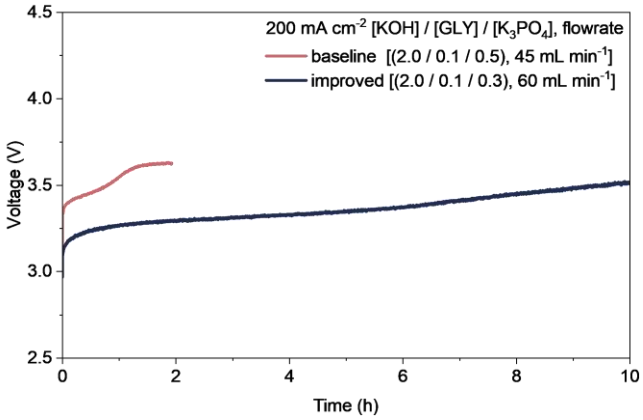

Figure S26. Full cell operating voltage over time at 200 mA cm^{-2} using the capture solvent and flow rate tailored for 100 mA cm^{-2} (baseline) and an optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.3 M K_3PO_4 operated at 60 mL min^{-1} (improved).

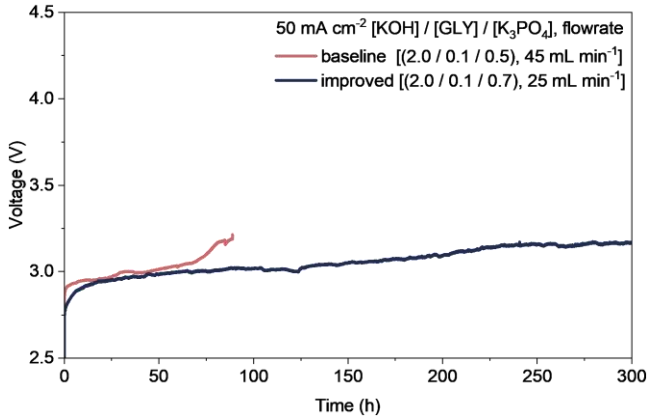

Figure S27. Full cell operating voltage over time at 50 mA cm^{-2} using the capture solvent and flow rate tailored for 100 mA cm^{-2} (baseline) and an optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.7 M K_3PO_4 operated at 25 mL min^{-1} (improved).

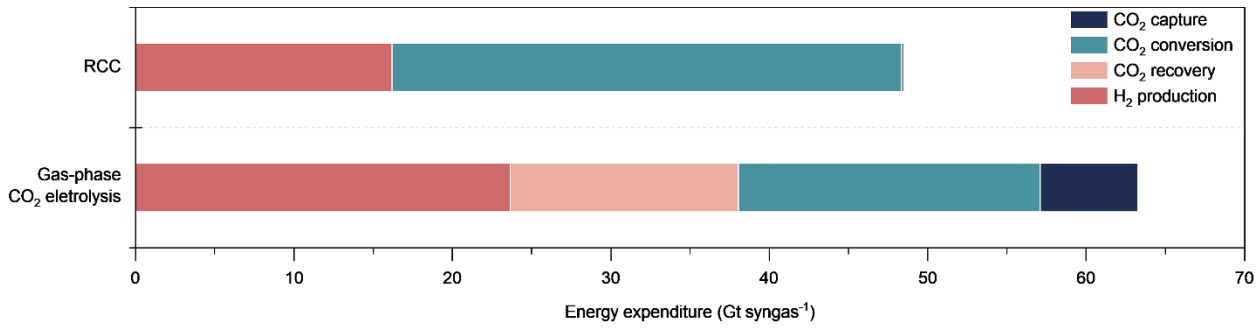

Figure S28. Summary of energy expenditure comparison between gas-phase CO_2 electrolysis and RCC.

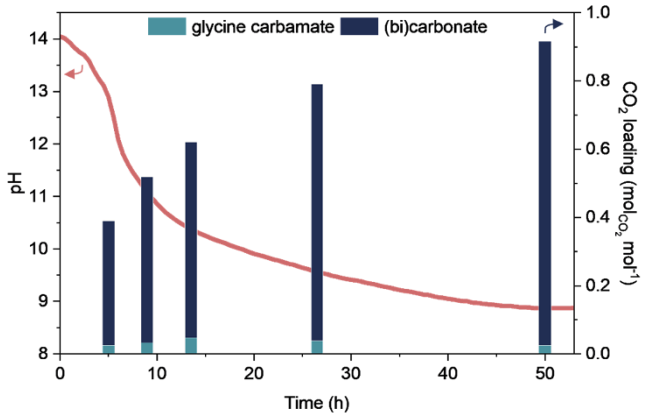

Figure S29. pH and CO_2 loading over time when purging the optimized capture solution, consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4, with 15 vol% CO_2 balanced with N_2. The chemisorbed CO_2 is in the form of glycine carbamate and (bi)carbonate.

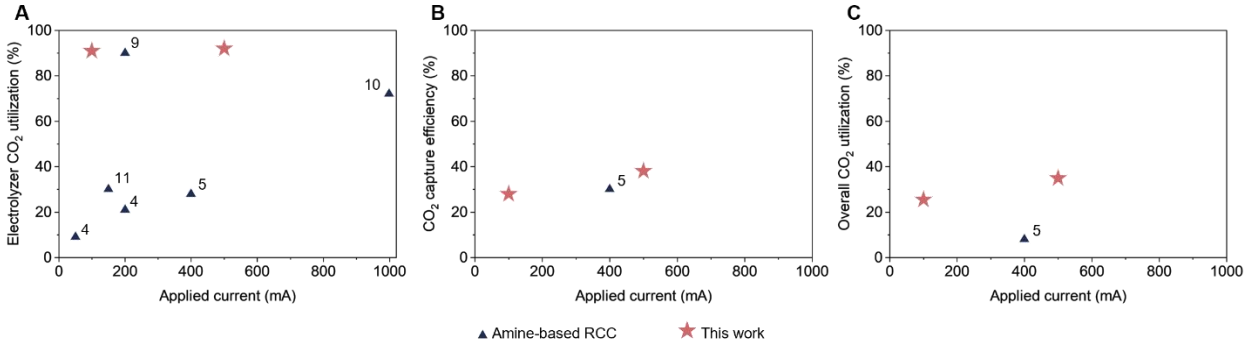

Figure S30. Comparing carbon efficiencies of state-of-the-art amine-based RCC versus total applied current density.[2,3,6] Details are tabulated in Table S4. (A) Electrolyzer CO_2 utilization. (B) CO_2 capture efficiency. (C) Overall CO_2 utilization.

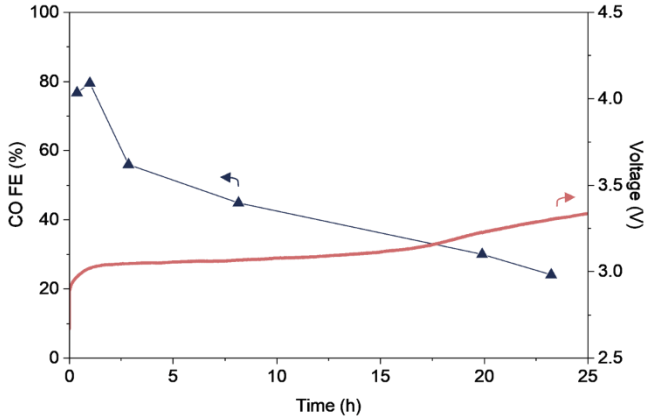

Figure S31. CO FE and non-iR compensated full cell operating voltage over time in a 5 cm^2 electrolyzer.

Table S1. Summary of state-of-the-art amine-based RCC for stability comparison.

Capture solvent	CO_2 capture	Post-capture species (pH)	Catalyst	Current density (mA cm^{-2})	CO partial current density (mA cm^{-2})	Stability (h)	Ref.
2 M KOH + 0.1 M GLY + 0.5 M K_3PO_4	continuous	carbamate & bicarbonate (9.0)	CoPc-CNT	100	60	75	*This work*
2 M KOH + 0.1 M GLY + 0.7 M K_3PO_4	continuous	carbamate & bicarbonate (8.7)	CoPc-CNT	50	30	250	*This work*
2 M K-GLY + 0.1 M KH_2PO_4	bulk	carbamate & bicarbonate (8.1)	Ni single atom catalyst (SAC)	50	32	32	2
1 M K_2CO_3 + 0.1 M K-GLY	continuous	carbamate & bicarbonate (8.9)	Ag NP	100	30	30	3
3 M triethylamine (TREA)	continuous	bicarbonate (7–8)	Ag-coral	100	35	70	1
5 M monoethanolamine (MEA)	bulk	carbamate	Ni SAC	50	32	10	4
1 M aminomethyl propanol (AMP)	continuous	bicarbonate (7)	Ag	11	10	8	5
5 M MEA + 2 M potassium chloride (KCl)	bulk	carbamate & bicarbonate (8.3)	Ag NP	50	36	3	6

Table S2. Comparison of AAS for stable RCC.

AAS	Amino acid molar mass (g mol^{-1})	pK_{a1} (α-carboxyl group)	pK_{a2} (α-amino group)	pK_{a3} (side chain)	pH after saturation with 15 vol% CO_2	Onset time of CO FE decline (h)
K-GLY	75.07	2.34	9.60		8.40	13
K-LYS	146.19	2.18	8.95	10.79	8.60	1
K-PRO	115.13	1.99	10.60		8.48	2
K-SER	105.09	2.21	9.15		8.21	3
K-SAR	89.09	2.06	10.35		8.46	5

Table S3. Key COMSOL model parameters

Parameters	Value	Unit	Reference
L_{CCL}	7	μm	*This work*
L_{CDL}	135	μm	*This work*
L_{CEM}	178	μm	*This work*
L_{ACL}	50	μm	*This work*
$P_{\mathrm{CO_2}}$	1	atm	9
$h_{\mathrm{G,0}}$	-0.0172	1	9
$\sigma_{\mathrm{CCL,electrolyte}}$	4.56	S m^{-1}	10
$\sigma_{\mathrm{CCL,electrode}}$	8×10^{4}	S m^{-1}	10
$\sigma_{\mathrm{ACL,elctrolyte}}$	4.56	S m^{-1}	10
$\sigma_{\mathrm{ACL,elctrode}}$	1.4×10^{7}	S m^{-1}	10
$\sigma_{\mathrm{CDL,elctrolyte}}$	4.56	S m^{-1}	10

σ_{CEM}	24.92	S m^{-1}	10
$\varepsilon_{\mathrm{CCL}}$	0.6	1	11
$\varepsilon_{\mathrm{ACL}}$	0.9	1	11
$\varepsilon_{\mathrm{CDL}}$	0.5	1	11
$\varepsilon_{\mathrm{CEM}}$	0.1	1	11
$D_{F,HCO3}$	1.19×10^{-9}	$m^2\ s^{-1}$	12
$D_{F,CO3}$	9.23×10^{-10}	$m^2\ s^{-1}$	12
$D_{F,OH}$	5.29×10^{-9}	$m^2\ s^{-1}$	12
$D_{F,H}$	9.31×10^{-9}	$m^2\ s^{-1}$	12
$D_{F,K}$	1.96×10^{-9}	$m^2\ s^{-1}$	12
$D_{F,CO2}$	1.91×10^{-9}	$m^2\ s^{-1}$	12
$D_{F,PO4}$	6.12×10^{-10}	$m^2\ s^{-1}$	13
$D_{F,HPO4}$	6.9×10^{-10}	$m^2\ s^{-1}$	13

$D_{F,H2PO4}$	6.05×10^{-10}	$m^2\ s^{-1}$	14
$D_{F,RNH3}$	1.32×10^{-9}	$m^2\ s^{-1}$	15
$D_{F,\mathrm{RNHCOO^-}}$	6.36×10^{-10}	$m^2\ s^{-1}$	16

Table S4. Cobalt concentration in the catholyte after 50 hours of electrolysis.

Catalyst	Co concentration (ppm)
CoPc-CNT with no Nafion	16.27
CoPc-CNT with 25 wt.% Nafion (Baseline)	1.42
CoPc-CNT with 33 wt.% Nafion	0.51
CoPc-CNT with 33 wt.% Nafion and EDTA (Confined)	0.38

Table S5. Summary of energy expenditure comparison between gas-phase CO_2 electrolysis and RCC.

Process	Gas-phase CO_2 electrolysis	RCC
CO_2 capture (GJ tsyngas^{-1})	6.18	0.14
CO_2 conversion (GJ tsyngas^{-1})	19.04	32.16
CO_2 recovery (GJ tsyngas^{-1})	14.41	0
H_2 production (GJ tsyngas^{-1})	23.64	16.19
Total (GJ tsyngas^{-1})	**63.27**	**48.49**

Table S6. Summary of state-of-the-art amine-based RCC for carbon efficiency comparison.

Ref.	Applied current (mA)	CO_2 capture efficiency (%)	Electrolyzer CO_2 utilization (%)	Overall CO_2 utilization (%)
This work (5 cm^2)	500	38	92	35
This work (1 cm^2)	100	28	91	25
2	50	N/R	9	N/R
2	200	N/R	21	N/R
3	400	30	28	8
6	200	N/R	90	N/R
7	998	N/R	72	N/R
8	150	N/R	30	N/R

Supplemental Note S1. Multi-physics modelling

The RCC system was modelled by COMSOL 6.3. This model aimed at understanding the cathode local pH while using different capture solvents.

Geometric model and assumptions

As shown in Figure S16, the one-dimensional model consisted of following domains: cathode catalyst layer (CCL), CO_2 diffusion layer (CDL), cation exchange membrane (CEM), and anode catalyst layer (ACL). The CO_2 was assumed to behave as an ideal gas, and the solubility depended on pressure, temperature, and salinity.

Governing equations and boundary conditions

The model considered the following processes: water dissociation reaction, carbonate equilibrium reactions, glycine equilibrium reactions, phosphate equilibrium reactions, electrochemical reactions, and charged species diffusion. Specific values can be found in Table S3.

The CO_2 concentration was calculated by Equation S1 using Henry's Law (Equation S2) and Sechenov Constant (Equation S3).

$$C_{\mathrm{CO_2,aq}} = P_{\mathrm{CO_2}} K_H^0 K_S \tag{S1}$$

$$\ln(K_H^0) = 93.4517 \times \frac{100}{T_0} - 60.2409 + 23.3585 \times \ln\left(\frac{T_0}{100}\right) \tag{S2}$$

$$K_S = 10^{-c_{\mathrm{K}}(0.0922+h_{G,0})-c_{\mathrm{OH}}(0.0839+h_{G,0})-c_{\mathrm{CO_3}}(0.1423+h_{G,0})-c_{\mathrm{HCO_3}}(0.0967+h_{G,0})} \tag{S3}$$

Where $C_{CO_2,aq}$, C_K, C_{OH}, C_{CO_3} and C_{HCO_3}represent the local concentration of dissolved CO_2, potassium (K^+), hydroxide ions (OH^-), carbonate ions (CO_3^{2-})and bicarbonate ions (HCO_3^-), P_{CO_2} is the partial pressure of CO_2, and $h_{G,0}$ is the Sechenov Constant for CO_2.

The electrochemical reactions of CO_2 to CO, hydrogen evolution reaction, and oxygen evolution reaction are listed below (Equations S4–S6):

$$CO_2 + H_2O + 2e^- \rightarrow CO + 2OH^- \tag{S4}$$

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^- \tag{S5}$$

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^- \tag{S6}$$

The water dissociation reaction is below (Equation S7):

$$H_2O \rightarrow H^+ + OH^- \tag{S7}$$

The carbonate equilibrium chemical reactions are as follows (Equations S8–S10):

$$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-} \tag{S8}$$

$$HCO_3^- + OH^- \rightleftharpoons CO_3^{2-} + H_2O \tag{S9}$$

$$CO_{2,aq} + OH^- \rightleftharpoons HCO_3^- \tag{S10}$$

The glycine equilibrium reactions are listed below (Equations S11–S13):

$$\mathrm{RNH_3^+} \rightleftharpoons \mathrm{RNH_2} + \mathrm{H^+} \tag{S11}$$

$$\mathrm{RNH_2} + \mathrm{CO_{2,aq}} + \mathrm{OH^-} \rightleftharpoons \mathrm{RNHCOO^-} + \mathrm{H_2O} \tag{S12}$$

$$\mathrm{RNHCOO^-} + \mathrm{H_2O} \rightleftharpoons \mathrm{RNH_2} + \mathrm{HCO_3^-} \tag{S13}$$

The phosphate equilibrium reactions are listed below (Equations S14 and S15):

$$\mathrm{HPO_4^{2-}} + \mathrm{OH^-} \rightleftharpoons \mathrm{PO_4^{3-}} + \mathrm{H_2O} \tag{S14}$$

$$\mathrm{H_2PO_4^-} + \mathrm{OH^-} \rightleftharpoons \mathrm{HPO_4^{2-}} + \mathrm{H_2O} \tag{S15}$$

Ohm's law (Equation S16) was applied in the cathode catalyst layer (CCL), cathode diffusion layer (CDL), cation exchange membrane (CEM), and anode catalyst layer (ACL).

$$i = -\sigma \frac{\partial \phi}{\partial x} \tag{S16}$$

Where i denotes the applied current density, σ is the electrical conductivity, ϕ represents the electric field potential, and x denotes the layer thickness.

The Poisson equation (Equation S17) was employed to account for electroneutrality and space-charge effects within the membrane. The electromigration of charged species

($H^+, OH^-, K^+, HCO_3^-, CO_3^{2-}, RNH_3^+, RNHCOO^-, H_2PO_4^-, HPO_4^{2-}, PO_4^{3-}$) was governed by the electrolyte-induced electric potential, electroneutrality constraints, and membrane space-charge effects.

$$\varepsilon_0 \varepsilon_r \frac{\partial \psi}{\partial x} = F \sum_i z_i \, c_i + \rho_{\mathrm{CEM}} \tag{S17}$$

$$V = \phi + \psi \tag{S18}$$

Where ε_0 and ε_r correspond to the vacuum permittivity (8.85×10^{-12} F m^{-1}) and the relative permittivity of water (80), respectively. ψ is the electric potential arising from electroneutrality constraints and membrane space-charge effects, z_i is the valence of species i, c_i denotes the concentration of species i, and ρ_{CEM} represents the fixed space-charge density of the membrane.

The electrochemical reactions rates (Equations S19–S21) were calculated using the Faradaic efficiency obtained from experiment results.

$$r_{\mathrm{CO}} = -\frac{i}{F}\left(\frac{\mathrm{FE_{CO}}}{2}\right)\frac{\varepsilon_{\mathrm{CCL}}}{L_{\mathrm{CCL}}} \tag{S19}$$

$$r_{\mathrm{OH^-}} = \frac{i}{F}\frac{\varepsilon_{\mathrm{CCL}}}{L_{\mathrm{CCL}}} \tag{S20}$$

$$r_{\mathrm{H^+}} = \frac{i}{F}\frac{\varepsilon_{\mathrm{ACL}}}{L_{\mathrm{ACL}}} \tag{S21}$$

Where r_{CO}, $r_{\mathrm{OH^-}}$, and $r_{\mathrm{H^+}}$ are the production rates of CO, $\mathrm{OH^-}$, and $\mathrm{H^+}$, $\varepsilon_{\mathrm{CCL}}$and $\varepsilon_{\mathrm{ACL}}$ are the porosity of the catalyst layer and the anode layer, and L_{CCL} and L_{ACL} are the thickness of the electrochemically active catalyst of cathode layer and anode layer.

In the reaction–diffusion framework, species transport (Equations S22 and S23) was described using the Nernst–Planck equations. The effective diffusivities of the species were calculated based on the Millington–Quirk relationships (Equations S24 and S25).

$$\frac{\partial C_i}{\partial t} + \frac{\partial J_i}{\partial x} + r_i = R_i \tag{S22}$$

$$J_i = -D_i \frac{\partial C_i}{\partial x} - \frac{z_i D_i}{RT_0} F C_i \frac{\partial V}{\partial x} \tag{S23}$$

$$\tau_{F,i} = \varepsilon_i^{-\frac{1}{3}} \tag{S24}$$

$$D_i = \frac{\varepsilon_i}{\tau_{F,i}} D_{F,i} \tag{S25}$$

Where r_i denotes the production rate of species i, and R_i represents the net contribution from equilibrium reactions. Here, J_i is the molar flux of species i, D_i is the effective diffusion coefficient of species i, $\tau_{F,i}$ is the fluid-phase tortuosity, and ε_i is the porosity of each layer. $D_{F,i}$ denotes the diffusion coefficient of species i in the bulk fluid.

At the left boundary of the CCL, post-capture solutions corresponding to the baseline and optimized (i.e., buffered) conditions were supplied. For the baseline and buffered cases, the electrolyte contained 2.0 M and 0.1 M glycine, respectively, with the speciation among $\mathrm{RNHCOO^-}$, $\mathrm{RNH_2}$, and $\mathrm{RNH_3^+}$ determined by post-capture equilibrium relations. For the baseline case, the electrolyte had a bulk pH of approximately 8.4 and contained 2.0 M $K^+$and 0.53 M total carbonate species ($\mathrm{HCO_3^-/CO_3^{2-}}$). For the buffered case, the electrolyte was maintained at a bulk pH of approximately 9.0 and contained 3.5 M $\mathrm{K^+}$ and 1.78 M total carbonate species, together with an additional 0.5 M phosphate buffer ($\mathrm{H_2PO_4^-/HPO_4^{2-}/PO_4^{3-}}$). All aqueous species distributions were determined from the corresponding equilibrium reactions at 298 K.

At the right boundary of ACL, a constant aqueous solution of 0.05 M sulfuric acid was supplied. The cathode potential was applied at the left boundary of the CCL, while the ground potential was imposed at the right boundary of the ACL. The pH versus distance from the cathode is shown in Figure S17.

Supplemental Note S2. Energy expenditure comparison between gas-phase CO_2 electrolysis and RCC.

We compared the energy expenditure to produce 1 tonne of syngas, with H_2:CO ratio of 2, via gas-phase CO_2 electrolysis and RCC. The processes considered in this analysis include CO_2 capture for the initial feedstock, CO_2 conversion in an electrolyzer, CO_2 recovery to separate excess CO_2 from the syngas product stream, and H_2 production to adjust the syngas ratio. The values were obtained from state-of-the-art reports of stable and scalable systems. The energy comparison is summarized in Figure S28 and Table S5.

CO_2 capture

It was assumed that 1.37 tonne of CO_2 was initially captured per 1 tonne of syngas. For gas-phase CO_2 electrolysis, CO_2 was captured from point-source emissions using aqueous monoethanolamine (MEA) absorption. This process comprises of CO_2 absorption via a column (0.10 GJ tCO_2^{-1}), steam regeneration (4.00 GJ tCO_2^{-1}), and gas compression (0.40 GJ tCO_2^{-1}), resulting in a CO_2 capture energy of 4.50 GJ tCO_2^{-1} or 6.18 GJ $tsyngas^{-1}$.[17] Gas pre-treatment and impurities scrubbing were not considered. For RCC, the post-capture solution was directly used as the electrolyzer input, requiring only the absorption process and a CO_2 capture energy of 0.10 GJ tCO_2^{-1} or 0.14 GJ $tsyngas^{-1}$.

Gas-phase CO_2 electrolysis

The CO_2 conversion energy calculation assumed 95% CO FE, 5% H_2 FE, and full cell voltage of 3 V, resulting in a CO energy efficiency (EE) of 42% (Equation S26).[18] The conversion energy, calculated using Equation S27,[11] was 21.79 GJ tCO^{-1} or 19.04 GJ $tsyngas^{-1}$.

$$\text{CO EE (\%)} = \frac{1.33\ \text{V}}{\text{full cell voltage (V)}} \times \text{CO FE (\%)} \tag{S26}$$

$$\text{CO}_2\ \text{conversion energy}\ \left(\frac{\text{GJ}}{\text{t}_{\text{CO}}}\right) = \frac{257\ \frac{\text{kJ}}{\text{mol}_{\text{CO}}}}{28\ \frac{\text{g}_{\text{CO}}}{\text{mol}_{\text{CO}}} \times \text{CO EE (\%)}} \tag{S27}$$

Excess CO_2 in the outlet gas stream was removed via MEA absorption. With a CO_2 utilization of 30%,[18,19] the CO_2 recovery energy required was 14.41 GJ tsyngas^{-1}.

RCC

For RCC, the CO_2 conversion energy was calculated using Equations S26 and S27. Using the average values obtained in the full-cycle configuration (Figure 4C, with 60% CO FE, 3.2 V full cell voltage, and 91% electrolyzer CO_2 utilization), the RCC system required 36.79 GJ tCO^{-1} or 32.16 GJ tsyngas^{-1}. CO_2 recovery was not required as the CO_2 concentration in the product syngas (~5 vol%) was below the downstream threshold.

H_2 production

H_2 production via proton exchange membrane electrolysis with an energy of 192.62 GJ tH_2^{-1} was considered for syngas ratio tuning.[20] H_2 produced as a side product of CO_2 electrolysis lowers the H_2 demand from the external water electrolyzers. Gas-phase CO_2 electrolysis requires an additional 23.64 GJ tsyngas^{-1} for H_2 production. RCC requires an additional 16.19 GJ tsyngas^{-1} for H_2 production.

REFERENCES

1. Langie, K.M.G., Tak, K., Kim, C., Lee, H.W., Park, K., Kim, D., Jung, W., Lee, C.W., Oh, H.-S., Lee, D.K., et al. (2022). Toward economical application of carbon capture and utilization technology with near-zero carbon emission. Nat Commun *13*, 7482. https://doi.org/10.1038/s41467-022-35239-9.

2. Xiao, Y.C., Sun, S.S., Zhao, Y., Miao, R.K., Fan, M., Lee, G., Chen, Y., Gabardo, C.M., Yu, Y., Qiu, C., et al. (2024). Reactive capture of CO2 via amino acid. Nat Commun *15*, 7849. https://doi.org/10.1038/s41467-024-51908-3.

3. Kim, Y., Lees, E.W., Donde, C., Jewlal, A.M.L., Waizenegger, C.E.B., De Hepcée, B.M.W., Simpson, G.L., Valji, A., and Berlinguette, C.P. (2024). Integrated CO2 capture and conversion to form syngas. Joule *8*, 3106–3125. https://doi.org/10.1016/j.joule.2024.10.010.

4. Kim, J.H., Jang, H., Bak, G., Choi, W., Yun, H., Lee, E., Kim, D., Kim, J., Lee, S.Y., and Hwang, Y.J. (2022). The insensitive cation effect on a single atom Ni catalyst allows selective electrochemical conversion of captured CO_2 in universal media. Energy Environ. Sci. *15*, 4301–4312. https://doi.org/10.1039/D2EE01825J.

5. Ahmad, N., Chen, Y., Wang, X., Sun, P., Bao, Y., and Xu, X. (2022). Highly efficient electrochemical upgrade of CO2 to CO using AMP capture solution as electrolyte. Renewable Energy *189*, 444–453. https://doi.org/10.1016/j.renene.2022.03.002.

6. Lee, G., Li, Y.C., Kim, J.-Y., Peng, T., Nam, D.-H., Sedighian Rasouli, A., Li, F., Luo, M., Ip, A.H., Joo, Y.-C., et al. (2021). Electrochemical upgrade of CO2 from amine capture solution. Nat Energy *6*, 46–53. https://doi.org/10.1038/s41560-020-00735-z.

7. Diaz, L.A., Gao, N., Adhikari, B., Lister, T.E., Dufek, E.J., and Wilson, A.D. (2018). Electrochemical production of syngas from CO_2 captured in switchable polarity solvents. Green Chem. *20*, 620–626. https://doi.org/10.1039/C7GC03069J.

8. Pérez-Gallent, E., Vankani, C., Sánchez-Martínez, C., Anastasopol, A., and Goetheer, E. (2021). Integrating CO_2 Capture with Electrochemical Conversion Using Amine-Based Capture Solvents as Electrolytes. Ind. Eng. Chem. Res. *60*, 4269–4278. https://doi.org/10.1021/acs.iecr.0c05848.

9. Weisenberger, S., and Schumpe, A. (1996). Estimation of gas solubilities in salt solutions at temperatures from 273 K to 363 K. AIChE J. *42*, 298–300. https://doi.org/10.1002/aic.690420130.

10. Gabardo, C.M., O'Brien, C.P., Edwards, J.P., McCallum, C., Xu, Y., Dinh, C.-T., Li, J., Sargent, E.H., and Sinton, D. (2019). Continuous Carbon Dioxide Electroreduction to Concentrated Multi-carbon Products Using a Membrane Electrode Assembly. Joule *3*, 2777–2791. https://doi.org/10.1016/j.joule.2019.07.021.

11. Xiao, Y.C., Gabardo, C.M., Liu, S., Lee, G., Zhao, Y., O'Brien, C.P., Miao, R.K., Xu, Y., Edwards, J.P., Fan, M., et al. (2023). Direct carbonate electrolysis into pure syngas. EES. Catal. *1*, 54–61. https://doi.org/10.1039/D2EY00046F.

12. Xu, Y., Edwards, J.P., Liu, S., Miao, R.K., Huang, J.E., Gabardo, C.M., O'Brien, C.P., Li, J., Sargent, E.H., and Sinton, D. (2021). Self-Cleaning CO_2 Reduction Systems: Unsteady Electrochemical Forcing Enables Stability. ACS Energy Lett. *6*, 809–815. https://doi.org/10.1021/acsenergylett.0c02401.

13. Parkhurst, D., and Appelo, C.A.J. (2013). Description of input and examples for PHREEQC version 3: A computer program for speciation, batch-reaction, one-dimensional transport, and inverse geochemical calculations https://doi.org/10.3133/tm6A43.

14. Zhang, H., Davison, W., Gadi, R., and Kobayashi, T. (1998). In situ measurement of dissolved phosphorus in natural waters using DGT. Analytica Chimica Acta *370*, 29–38. https://doi.org/10.1016/S0003-2670(98)00250-5.

15. Khanal, S.P., Kandel, Y.P., and Adhikari, N.P. (2019). Transport properties of zwitterion glycine, diglycine, and triglycine in water. AIP Advances *9*, 065303. https://doi.org/10.1063/1.5099069.

16. Ma, Y., Zhu, C., Ma, P., and Yu, K.T. (2005). Studies on the Diffusion Coefficients of Amino Acids in Aqueous Solutions. J. Chem. Eng. Data *50*, 1192–1196. https://doi.org/10.1021/je049582g.

17. Li, K., Leigh, W., Feron, P., Yu, H., and Tade, M. (2016). Systematic study of aqueous monoethanolamine (MEA)-based CO2 capture process: Techno-economic assessment of the MEA process and its improvements. Applied Energy *165*, 648–659. https://doi.org/10.1016/j.apenergy.2015.12.109.

18. Moreno-Gonzalez, M., Berger, A., Borsboom-Hanson, T., and Mérida, W. (2021). Carbon-neutral fuels and chemicals: Economic analysis of renewable syngas pathways via CO2 electrolysis. Energy Conversion and Management *244*, 114452. https://doi.org/10.1016/j.enconman.2021.114452.

19. Alerte, T., Edwards, J.P., Gabardo, C.M., O'Brien, C.P., Gaona, A., Wicks, J., Obradović, A., Sarkar, A., Jaffer, S.A., MacLean, H.L., et al. (2021). Downstream of the CO_2 Electrolyzer: Assessing the Energy Intensity of Product Separation. ACS Energy Lett. *6*, 4405–4412. https://doi.org/10.1021/acsenergylett.1c02263.

20. Jang, D., Kim, J., Kim, D., Han, W.-B., and Kang, S. (2022). Techno-economic analysis and Monte Carlo simulation of green hydrogen production technology through various water electrolysis technologies. Energy Conversion and Management *258*, 115499. https://doi.org/10.1016/j.enconman.2022.115499.